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Plasma driven superpermeation of hydrogen through group Va metals

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A barrier for the absorption of hydrogen molecules generated by a nonmetallic film on a metal surface may result in superpermeation of suprathermal hydrogen particles. Virtually all the sticking or implanted particles may permeate through the metallic membrane, with a permeation flux depending neither on the metal temperature nor on its thickness. Niobium and vanadium membranes with oxicarbide monolayer films were immersed in a hydrogen plasma in order to study their interaction with hydrogen ions in the energy range from a few eV to 250 eV, controlled by changing the membrane bias. A stable superpermeation flux was observed up to ion energies of tens of eV without special efforts to maintain the surface film. We found that the nonmetal monolayer film remains absolutely stable under the sputtering, including chemical sputtering, by hydrogen particles of energy less than $\approx$50 eV. Thus superpermeability appears to be an intrinsic feature of the investigated systems of ''nonmetallic film–metal–hydrogen particles of energy lower than 50 eV.''
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I. INTRODUCTION

A metallic membrane of macroscopic thickness may be superpermeable to hydrogen particles with energy (kinetic, internal, or chemical) exceeding $\sim$1 eV.[1,2] This means that virtually the whole flux of suprathermal particles implanted in the metal lattice, or just sticking to its surface, passes through the membrane, with the permeation flux depending neither on the membrane temperature nor on its thickness. When the incident energy is 1 keV or higher, the implantation probability is close to unity. Therefore almost the whole incident flux permeates through the membrane and the permeability of a solid membrane approaches in this case its theoretical limit. At lower energies, a fraction of the incident suprathermal particles can be reflected back from the surface;[3] still the probability of implantation (or of sticking) stays comparable to unity even in this case, as also does the permeation probability. Since the implantation/sticking probability for suprathermal particles does not depend on the metal temperature, the permeation flux should also be temperature independent at superpermeability, a characteristic of the phenomenon which is actually observed.

The underlying physical ideas are illustrated on the potential diagram of the hydrogen–metal membrane system (Fig. 1). A high surface barrier, E_{r0}, is most important here. This barrier prevents the dissociative absorption of thermal hydrogen molecules and makes the membrane virtually impermeable to molecules. Suprathermal hydrogen particles (atoms, ions, molecules) with the total energy higher than E_{r0} can pass into the metal despite this barrier. This energy may be not only kinetic. For example, free hydrogen atoms having thermal kinetic energy can be considered as suprathermal particles due to their excess of chemical energy (2.2 eV with respect to an atom in a H_2 molecule, see Fig. 1). The particles ''cool down'' in the metal lattice, equilibrate in the state of solid solution (as atoms) and are being ''reflected'' back by the $E_{r0}-E_b$ barrier as they attempt to escape by means of the thermal re-emission through recombination with another absorbed atom. If such a reflection is strong enough (the ''strong reflection condition'' [1,4]), the dissolved atoms may reach the opposite side of the membrane by diffusion with a higher probability than that of being re-emitted back. The downstream barrier $E_{rL}-E_b$ also reflects the absorbed atoms, and thus the atoms ''oscillate'' by means of random-walk diffusion between the two boundaries. As a result, the absorbed atoms ''forget'' their starting position and are being desorbed on either of the two sides of the membrane depending only on the relative heights of the corresponding surface barriers. In particular, if $E_{r0}>E_{rL}$, a major part of absorbed particles will pass through the membrane independently of its temperature and thickness. Two important features of superpermeation follow from the above:

(a) The characteristic time of permeation flux establishment (τ) is long compared to the time of one diffusion passage across the membrane (τ_D), $\tau \gg \tau_D$. This is the result of repeated crossing of the membrane before desorption.
(b) The permeating hydrogen can be automatically compressed by orders of magnitude, i.e., it can pass through the membrane from the lower pressure side to the side where the hydrogen pressure is higher by several orders. This is related to the fact that the absorbed atoms are thermally desorbed from the metal at its moderate temperature ($<$1000 °C) only when associated into molecules and that the superpermeable membrane is virtually impermeable to thermal molecules.

It is the above strong reflection condition that usually puts the limits on the incident flux density, j_i, membrane

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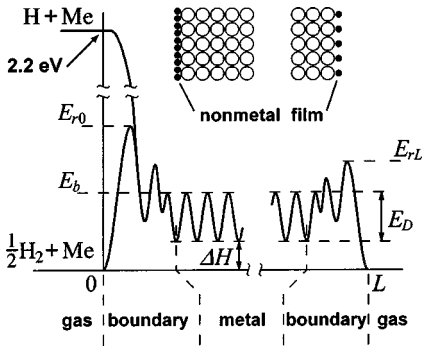


FIG. 1. Potential diagram of a hydrogen-metal system. The energy of a hydrogen atom in a hydrogen molecule is taken as the zero energy level and correspondingly, the energy of a free atom equals 2.2 eV. ΔH is the level of the atom in the state of solution (the enthalpy of solution), E_D is the activation energy of diffusion, and $E_b=\Delta H+E_D$. In case of Group V metals, both ΔH and $E_b<0$.

thickness, L, and on the temperature range at which superpermeability is possible.[1,2,4] Still this condition is actually always satisfied in the case of the Group Va metals (V, Nb, and Ta)[5] due to their lowest (among all metals) level of the fundamental bulk barrier, E_b, $(E_b<0)$[6] combined with a usually high[2,7,8] surface barrier E_r. This is the reason why the Group Va metals appear to be the most promising for superpermeable membranes, and chosen as the object of the present investigation.

Taking into account these peculiarities of the hydrogen–Group V metal systems, the permeation flux density j_p may be expressed as:[1,2,4]

$$j_p=\sum \xi_k j_{ik} \frac{\alpha_L}{(\alpha_0+\alpha_L)}, \tag{1}$$

where j_{ik} is the incident flux of the suprathermal hydrogen particles of type k, ξ_k is the probability for the k-type particles to be absorbed at one impact on the upstream surface, α_0 and α_L are the ordinary sticking coefficients into the state of dissociative adsorption of thermal H_2 molecules for the up- and downstream sides correspondingly, associated with the surface barriers (Fig. 1) as follows:[1]

$$\alpha_0=\alpha_0\sigma_0 \exp(-2E_{r0}/RT);$$
$$\alpha_L=\alpha_L\sigma_L \exp(-2E_{rL}/RT). \tag{2}$$

Here σ_0 and σ_L are the parts of the surface occupied by the centers of thermal absorption/desorption characterized by E_{r0} and E_{rL} barriers, respectively, while the remaining part of the surface (i.e., $1-\sigma$) is blocked by a substantially higher barrier and is not active in the thermal absorption/desorption. α_0 and α_L are pre-exponential factors ranging typically between 0.1 and 1. The sticking coefficients are related to the often employed (e.g., Refs. 10 and 11) rate constants of recombinative release, k_r, via:

$$\alpha_0=k_{r0}S^2/Z; \quad \alpha_L=k_{rL}S^2/Z, \tag{3}$$

where S is the hydrogen solubility constant,[6] and $Z=(2\pi mkT)^{1/2}$ is the coefficient determined from the kinetic theory, with m as the molecular mass and k as the Boltzmann's constant.

According to the above definition and Eq. (1), a membrane will be superpermeable to suprathermal particles, if their passage into the metal does not depend on E_{r0}, σ_0, and T, with the implantation probability constrained only by the reflection coefficient, β_k (i.e., with ξ_k ranging between 0.1 and 1, depending on the particle energy):

$$\xi_k=1-\beta_k=\text{const}(T), \tag{4}$$

and if the *thermal* processes of dissociative absorption/recombinative desorption are blocked to a larger extent at the upstream rather than at the downstream surface:

$$\alpha_0\leqslant\alpha_L. \tag{5}$$

Typically, monolayers of nonmetallic elements (a ''real'' surface) are responsible for the E_r barrier and small σ [i.e., for small α in accordance with Eq. (2)].[1,2,4,5,7,8,10–16] It is important to note that the state of a metal surface covered with nonmetallic monoatomic films is, as a rule, a great deal more chemically and thermally stable than the corresponding bulk compounds. Moreover, a metal surface covered with a nonmetal monolayer may be a thermodynamically stable state of the surface under vacuum conditions, i.e., there are mechanisms that can maintain automatically such a monolayer coverage and withstand perturbations, if the metal temperature is high enough to provide for sufficient rates of recovering mechanisms like surface segregation, dissolution and desorption.

It should be noted that, despite these arguments in favor of the possible stability of the monolayer film, the asymmetry condition (5) appears to be very severe, since energetic hydrogen particles are incident on the upstream side only, whereas the downstream side is not subject to anything similar. Furthermore one should not necessarily remove the monolayer film from the upstream surface to violate the condition (5): α_0 (and k_{r0}) may increase by orders of magnitude due to a relatively small concentration of film defects which may serve as centers to facilitate desorption [i.e., with a lower E_{r0} and higher σ_0 in Eq. (2)].

Previous experiments were performed using two types of hydrogen particles: thermal atoms and accelerated ions. Superpermeability to thermal hydrogen atoms was observed on membranes of Ni,[12] Pd,[12–14] Fe,[11,15] and Nb[2,5,16,17] in combination with impurity monolayers of O, C, and S, either deposited in a controlled way or of natural origin. Two main experimental methods were employed: the atomic beam techniques[12,15] and the dissociation at an incandescent body.[2,5,10,13,14,16,17] There are two nontrivial points regarding the superpermeation of thermal atoms.

First, the unimpeded (i.e., ζ has a value between 0.1 and 1, which does not depend on T) passage of thermal atoms through nonmetallic monolayers is clearly indicated by the very fact of superpermeability.

Second, the superpermeation remains absolutely stable with all the investigated ''metal–nonmetallic impurity'' combinations and membrane temperatures, with no special efforts taken to maintain the nonmetallic monolayers (the ''spontaneous'' superpermeability), even though the ability of hydrogen atoms to chemically sputter nonmetals is well known (J. Roth in Ref. 18). Note that we do not refer here to the above mentioned case of a nonmetallic monolayer exist-

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ing under sputtering due to its automatic maintenance: thermal atoms do not sputter nonmetallic monolayers *at all* in the systems investigated. In particular, an initially symmetric membrane ($\alpha_0 = \alpha_L$) always remains symmetric under the action of thermal atoms even at a temperature as low as 20 °C (i.e., with the surface segregation ''frozen'') and under ultrahigh vacuum (UHV) conditions (i.e., with no impurity replenishment from the gas phase).[13,14]

The pattern of behavior differs sharply in the case of permeation of accelerated ions (ion beam techniques, $E > 70$ eV/proton): despite a guaranteed deep implantation, the initially high permeability falls by orders of magnitude under the action of fast ions under UHV conditions because of the physical sputtering of impurities (the so called ''permeation spike,'' see e.g., Refs. 2, 19). Usually such a drop in permeation is the result of the creation of defects (holes) in the impurity monolayer that is dynamically maintained due to the impurity surface segregation (Ref. 7 and G. Betz and G. K. Wehner in Ref. 18). As a result, α_0 increases by orders of magnitude, while still remaining much smaller than 0.1 (the value that one may regard as the lowest typical for a clean metallic surface[20]).

Thus the spontaneous superpermeation (as in the case of thermal atoms) is not observed here. Special efforts are needed to maintain the required state of the surface, and in particular, the condition (5), under ion sputtering. For instance, admitting chemically active gases (O_2, H_2O, CH_4, H_2S) onto the membrane upstream side in the course of ion bombardment increases the extremely small steady-state ion driven permeation up to superpermeation, and the latter exists as long as the chemically active gas is present.[2,7,17,21]

As it follows from above, it is the physical sputtering that results in a significant difference in the superpermeation driven by fast ions and by thermal atoms. Therefore it is of particular interest to investigate the behavior of hydrogen particles whose energy is higher than that of thermal atoms though still below the physical sputtering threshold. This refers to the energy range of eV to tens of eV. While it is virtually impossible to get particles in this energy range with ion beam techniques, it can be done easily with the use of cold hydrogen plasma with additional control of incident ion energy by membrane bias.

Several published papers refer to the plasma driven permeation of hydrogen through metallic membranes: experiments by Günterschulze *et al.*[22] and by Pillath *et al.*[23] with Fe membranes, and those by Park *et al.*[24] with a Pd membrane, where a rather high plasma driven permeation was observed. Our purpose is to demonstrate the possibility of the plasma driven superpermeation using the most suitable membrane materials (Group Va metals vanadium and niobium), focussing our attention on the particular role played by the surface facing the plasma and its modifications under the action of hydrogen particles of controlled energy. Furthermore plasma driven superpermeation may be particularly important for applications in fusion as well as in plasma-chemical technologies.

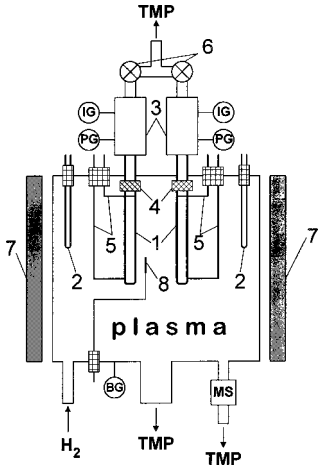


FIG. 2. Schematic of the plasma-membrane setup; (1) membranes, (2) filaments, (3) downstream vacuum chambers, (4) insulators, (5) heavy-current leads, (6) valves, (7) multicusp magnet system, (8) electrostatic probe. IG—ionization gauge, PG—Pirani gauge, BG—Baratron gauge, MS—quadrupole mass spectrometer, TMP—turbomolecular pump.

II. EXPERIMENTAL SETUP

A uniform hydrogen plasma filling a chamber 44 cm in diameter and 45 cm in height is generated by the electric discharge between a set of sixteen hot tantalum cathode filaments (0.5 mm in diameter, with a total area 30 cm^2) located close to the chamber wall in a multicusp magnetic field and the chamber wall, serving as an anode as shown in Fig. 2.[25] The plasma density is about 5×10^{10} cm^{-3}, and the electron temperature is approximately 0.6 eV[25] under the typical conditions of this experiment (discharge current 25–30 A, discharge voltage of 60 V, gas pressure 1–3 mTorr).

Two turbo-molecular pumps (50 ℓ/s each) continuously operate in the down- and upstream chambers. The UHV vacuum chamber on the downstream side was bakeable while the plasma chamber on the upstream side was not bakeable. The evaporation of Ta from the filament cathodes onto the chamber wall in the course of plasma operation, as well as wall conditioning by the plasma, contributed to reducing the pressure of gas impurities. As a result, the total pressure of impurities was in the order of 10^{-6} Torr during plasma operation, and a quadrupole mass spectrometer indicated that CO was the main impurity, and H_2O the next one (2×10^{-7} Torr). The pressure in the downstream chamber is measured with Bayard–Alpert, Pirani, and Baracell gauges.

Several suprathermal hydrogen particle species are generated in the discharge determining the sum in Eq. (1).[26] Among them are H^+, H_2^+, H_3^+, and H^- ions. Among the positive ions H_2^+ and H_3^+ are the predominant species.[27] This has to be mentioned because the energy per proton depends on the ion species. The ion temperature is a few tenths of an eV. Positive ions can be accelerated to the membrane to energies up to hundreds of eV by biasing the membrane, which is insulated from the walls by a ceramic. The density of hydrogen atoms is typically a few percent and greatly exceeds the ion density (the rest are hydrogen molecules).[26,28] The temperature of the atoms is a few tenths eV (Ref. 26) but energetic atoms generated by the primary

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energetic electrons in the near-wall region might also reach the membrane in our pressure range. The density of vibrationally excited molecules ($v''=1$) is comparable to that of atoms,[26] but their total energy is probably not sufficient for passing the surface barrier. Thus hydrogen atoms are expected to give the main input into the incident flux of suitable suprathermal particles while ions play the important role of particles whose energy may be controlled and varied over a wide range.

Two Ohmically heated tubular membranes of Nb and V (1 cm in diameter, 18 cm long, 0.01 cm wall thickness) are immersed into the plasma. Auger electron spectroscopy (AES) analysis of the membrane material showed that oxygen and carbon were the main nonmetallic impurities that remained, in the form of a monolayer coating, on the surface of both specimens after their heating (>550 °C) in vacuum. The same impurities segregated again to the surface from the metal bulk after the surface was submitted to Ar^+ ion bombardment followed with heating in UHV.

The AES analysis of the membrane upstream surfaces after their long-term operation in our hydrogen plasma yielded the same results, but in this case both oxygen and carbon were discovered located on the top of the tantalum film evaporated from the filaments and covering the Nb and V. Note that the three metals (V, Nb, and Ta) belong to the Group Va and their chemical properties, particularly the characteristics of their interaction with hydrogen, are very similar.[6] It was shown in earlier experiments on the superpermeability of Nb to thermal hydrogen atoms[5,16,17] that the deposition of Ta (like that of fresh Nb) does not affect the basic features of the behavior of the Nb membrane.

III. EXPERIMENTAL RESULTS AND DISCUSSION

A. Evidence for superpermeation

The experiment with a vanadium membrane presented in Fig. 3 is typical for both vanadium and niobium. The permeation driven by thermal molecules, thermal atoms, and plasma is observed successively. The incandescent tantalum filaments were used first to dissociate molecular hydrogen, allowing us to examine whether our membranes are actually superpermeable to thermal hydrogen atoms.[5,16,17] Switching on the discharge in addition to the filaments allowed us to observe the behavior of such an initially superpermeable membrane in the plasma.

The upstream pressure is kept constant (1.6 mTorr) due to the continuous flow of H_2, and the downstream pressure, P_c, is determined by the permeation flux, j_p, ($P_c \propto j_p$ with a constant chamber pumping speed; the latter was $\approx 2\ l/s$ for H_2 in our case).

In spite of the high membrane temperature, T, (800 °C), the permeation due to hydrogen molecules is very small (the corresponding pressure increment, P_c, is about 0.07 mTorr and is not visible on the scale of Fig. 3), as it usually occurs when a metal surface is passivated with a nonmetallic film.[2]

Switching on only the filaments (2000–2100 °C) results in an increase of the permeation flux by more than an order of magnitude. Although we do not know the value of the incident H atom flux, j_i, (and, correspondingly, of j_p/j_i) with a proper accuracy here, there is evidence that we are dealing with a typical atom driven superpermeation.

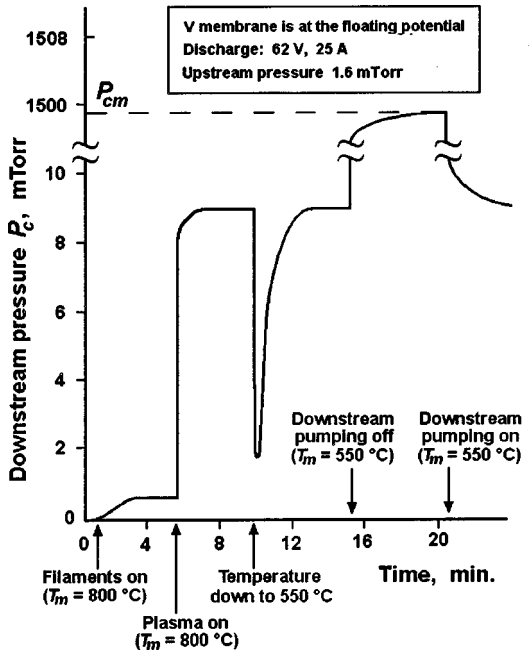


FIG. 3. Typical permeation experiment demonstrating the plasma driven superpermeation. The pumping speed on the downstream side is $\approx 2\ l/s$ for H_2. When the downstream pumping is on, $j_p \approx 3\times 10^{18} \times P_c$ atom/cm^{2}s, where P_c is in Torr.

Then the discharge (25 A, 62 V) is switched on with the membrane at floating potential. The permeation flux density rises more than ten times. The next step consists in decreasing the membrane temperature from 800 to 550 °C during the plasma operation. After a transient process connected with the accumulation in the bulk of the metal of the hydrogen being absorbed, the permeation flux comes back to exactly the former level. Thus the temperature independence of permeation is observed, which is one of the most characteristic features of superpermeation.[1,2,4,7,11–16] Several other evidences in favor of plasma driven superpermeation are: (a) $\tau \gg \tau_D$ is observed; (b) the permeation flux, j_p, equals 0.1–0.3 of the incident flux, j_i, of hydrogen atoms generated in the plasma [with the accuracy with which j_i is known (Sec. II)], which is in reasonable agreement with the expected implantation coefficient, ξ, for hydrogen atoms with a kinetic energy of a few tenths of eV.

Then the downstream pumping is cut off (by closing a valve—see Fig. 2). The downstream pressure rises to reach a maximum level, $P_{\rm cm}$, corresponding to the balance of the back permeation flux (driven by molecular hydrogen) and the direct plasma driven flux. With this balance reached, the ratio of down- and upstream pressures corresponds to a maximum degree of compression, q, attainable at given membrane temperature, plasma parameters, and state of membrane surface[1] ($q \approx 10^3$ in the case presented in Fig. 3).

Finally, the downstream pumping is switched on again, and the downstream pressure establishes in the end at the same level as before closing the valve. Thus the superpermeation regime of membrane operation remains quite stable during the whole experiment, with no particular measures

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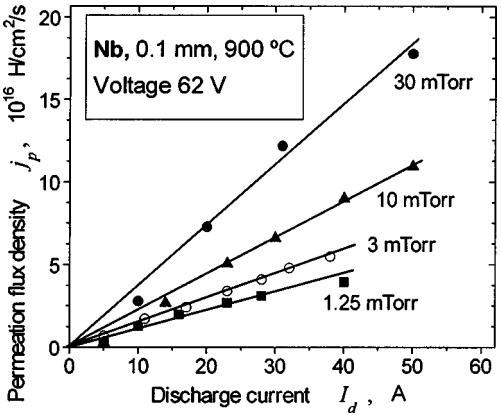


FIG. 4. Dependence of the permeation flux density on discharge current at different values of the hydrogen pressure on the upstream side. The permeation flux due to the dissociation on filaments is subtracted.

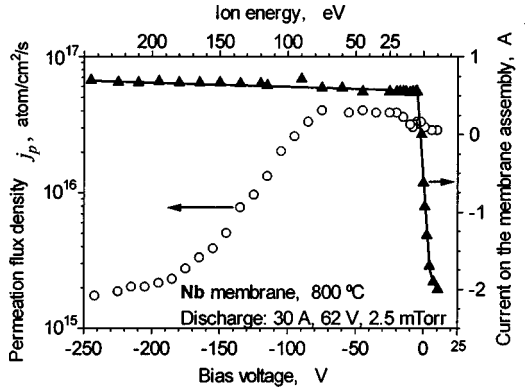


FIG. 5. Permeation flux density and current to the membrane assembly vs membrane bias.

taken to maintain it. Moreover, a stable plasma driven superpermeation of both the Nb and V membranes was observed during the whole experimental campaign (~300 hs) over the investigated temperature range (550–1000 °C).

B. Dependence of permeation flux on discharge current

The flux of primary electrons, i.e., the discharge current J_d, is responsible for the generation of suprathermal particles. Figure 4 shows that the proportionality of j_p to J_d is observed over the whole range of j_p and upstream pressures investigated. A standard examination of every point on the plots in Fig. 4 (experimental runs like the one presented in Fig. 3 that showed a stable and temperature independent permeation) confirms that we actually have the superpermeation regime. Thus the surface facing the plasma turns out to be capable of keeping up the state that gives rise to favorable asymmetry ($\alpha_0 < \alpha_L$) at all the investigated flux densities, in spite of the surface being acted upon by suprathermal particles.

One should note that the j_p values presented here ($>10^{17}$ cm^{-2} s^{-1}) exceed by orders of magnitude those reached with particle beam techniques. Thus a substantial progress has been achieved due to the use of plasma as a generator of energetic hydrogen. However technical reasons did not permit progression further up to higher fluxes even in this case, though the theoretical limit of j_p amounts to $\approx 10^{19}$ cm^{-2} s^{-1} for the superpermeation regime.[5,16]

C. Effect of membrane bias

Unlike in ion beam experiments[2,7,17,19,21] (with ion energy $E_k > 70$ eV/proton) where a conspicuous permeation drop due to the surface impurity sputtering with ions of 70 eV or higher kinetic energy was observed, we see a stable plasma driven superpermeation at the membrane floating potential (≈ -3 V with respect to plasma potential), in spite of the action of ions with kinetic energy of a few eV. Note that the energy released at the surface region at ion recombination could substantially exceed the kinetic energy of the incident ions. For example, for the H_2^+ ions it is [15.4+(0.4 $\times$2)]eV/ion, where 15.4 eV is the ionization potential of H_2^+ and 0.4 eV is the dissolution enthalpy in the hydrogen-Nb system (ΔH on Fig. 1).

By applying to the membrane a bias with respect to the plasma, one can investigate a wide range of kinetic energies of incident ions: from the lowest energies inaccessible by other means up to those typical for ion beam experiments. In particular, one may hope to detect the threshold energy at which the spontaneous superpermeability starts to be destroyed.

It should be noted that with changing the bias voltage we vary only the ion energy, whereas in our plasma the incident flux of energetic particles consists mainly of H atoms (Sec. II A). Correspondingly, changes in ion energy should manifest themselves mainly in the effects of ions on the state of membrane surface, with a resulting change in the permeation mainly driven by energetic neutrals of constant energy.

A Langmuir probe installed near the membrane (at $\approx$0.5 cm) showed that the bias has no effect on the plasma characteristics and on the plasma potential (relative to ground) over the whole range of bias voltages investigated (from +20 to −250 V).

The dependence of the permeation flux density, j_p, on bias potential, U_b, is presented in Fig. 5. Also presented is the current to the membrane assembly. The ion branch of this current–voltage plot reveals that the ion flux to the membrane is practically constant over this voltage range: a small current increase following the increase of negative bias may be attributed to the secondary ion-electron emission.

1. Contribution of ions

One can see in Fig. 5 the effect of the cutoff of ions (due to the positive bias) on the permeation flux density, j_p. This effect is not quite clearly expressed because a small peculiarity in the j_p behavior superimposes in the vicinity of $U_b \approx 0$. The smallness of the effect of ion cutoff is not in disagreement with our expectation, since the contribution of ions into the incident flux density does not exceed 10% of the density of the H atom flux (Sec. II), even if one takes into account that every H_2^+ and H_3^+ ion is equivalent to two or three H atoms, respectively.

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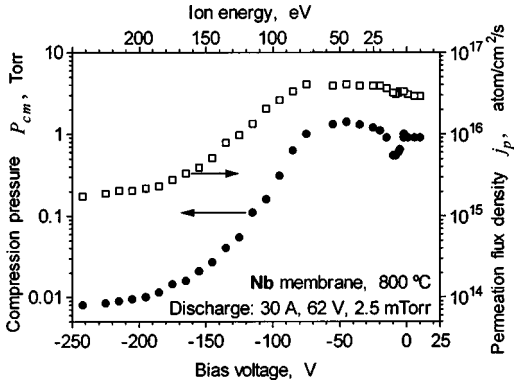


FIG. 6. Dependence of the maximum compression pressure beyond the membrane, $P_{\rm cm}$, on bias. The permeation flux density is presented here as well for comparison.

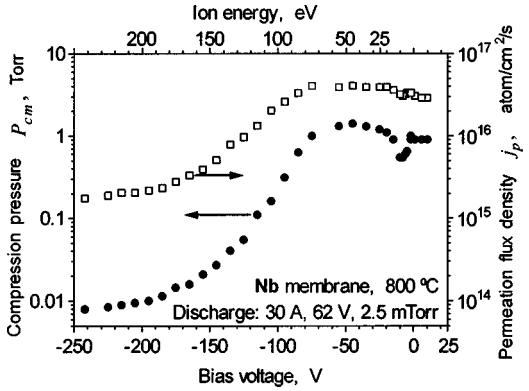


FIG. 7. Dependencies on membrane bias of the H_2 molecule sticking coefficient on the upstream side, α_0, and of the membrane asymmetry, α_L/α_0. To plot α_L/α_0 an α_L value (which is bias independent) was experimentally determined to be 1.2×10^{-4}. The rate constant of recombinative release, k_{r0}, can be found from α_0 by Eq. (3).

2. Plasma driven permeation

The permeation flux density, j_p, remains constant, in the first approximation, over the bias range of $0>U_b>-(60-70)$ V, although small j_p changes are observed here despite the constant incident flux. Thus a stable (spontaneous) plasma driven superpermeation that has been found at the floating potential is maintained at ion energies of up to 60–70 eV.

3. State of the upstream surface, compression

The observed approximate independence of the permeation flux on the bias potential in the mentioned range does not prove that the state of the surface facing the plasma remains unchanged throughout this range. According to Eq. (1), the permeation flux j_p may be rather insensitive to the state of this surface for the case of superpermeation with a favorable asymmetry ($\alpha_0<\alpha_L$) until the surface modification becomes so marked that it brings about a complete degradation of the favorable asymmetry (i.e., $\alpha_0\geqslant\alpha_L$).

However, the membrane ability to compress the permeating hydrogen (Sec. III A) is always very sensitive to any upstream surface changes. The back permeation flux caused by the thermal molecules accumulated in the downstream chamber is $ZP_c\ \alpha_0\alpha_L/(\alpha_0+\alpha_L)$, where Z is the kinetic theory coefficient at room temperature, and P_c is the downstream pressure measured with a gauge at room temperature. The maximum compression pressure beyond the membrane $P_{\rm cm}$ can be calculated from the balance of the direct plasma driven permeation flux, Eq. (1), and the backstream ''molecular'' permeation flux:

$$P_{\rm cm}=1/Z\sum\ \xi_k\ j_{ik}\times 1/\alpha_0. \tag{6}$$

Thus $P_{\rm cm}\propto 1/\alpha_0$ is valid at a constant implantation flux $\Sigma\xi_k\ j_{ik}$ (a conjecture $\Sigma\xi_k\ j_{ik}\approx$const looks quite reasonable at $U_b<0$, since neither the energy nor the flux of energetic neutrals depend on bias).

The dependence of $P_{\rm cm}$ on bias is presented in Fig. 6, along with that of the permeation flux j_p from Fig. 5. The dependencies on bias of the sticking coefficient, α_0, for thermal H_2 molecules and of the membrane asymmetry α_L/α_0, found from the presented data for $P_{\rm cm}$ and j_p with Eq. (6), are shown in Fig. 7. Note that the α_0 and α_L/α_0 values found here directly from the plasma experiment refer to the state of the upstream surface that exists under exposure to plasma.

One can see that all the effects of bias are more pronounced in the compression data, and therefore the ion energy has more influence on the state of the surface (i.e., on α_0) than it might appear only from the j_p dependence on bias. Still the overall patterns of bias dependencies remain basically similar in the first approximation: α_0 exhibits no substantial change up to an ion energy, E_k, of about 50 eV, and then sharply increases at energies exceeding this value, which one reasonably might attribute to the physical sputtering threshold for a monolayer film of light impurities (O and C) chemically bonded at the metal surface (sputtering of Nb and Ta requires a substantially higher energy) (see article by Betz and Wehner in Ref. 18).

This means that we see the effect of sputtering right when it actually starts, viz., one may conclude that, most probably, there is no sputtering at all (and no chemical sputtering as well) of the impurity monolayer at $E_k<50$ eV (just as with the thermal atoms). If this is true, an implication is that the observed spontaneous superpermeation is an intrinsic feature of the investigated system over this specific ion energy range and that it may exist for an unlimited time, depending neither on the inventory of impurities in the bulk metal nor on their presence in gas phase.

As we already noted above, a dramatic rise of α_0, and an ensuing superpermeability degradation similar to the one detected here at $E_k>50$ eV, were observed in ion beam experiments with a Nb membrane at $E_k>70$ eV/proton under UHV conditions.[7,17,21] Still the presence in the gas phase of oxygen- or carbon-containing impurities (at pressures of 10^{-6} Torr or higher at the ion beam density of 2×10^{15} proton cm^{-2} s^{-1}) resulted in that a low value of α_0 (i.e., $\alpha_0\leqslant\alpha_L$), and hence the conservation of superpermeability despite sputtering. Thus the very fact of the sharp α_0 increase at $E_k>50$ eV is an evidence that gaseous impurities in the given plasma-membrane experiment are reduced to such an extent (due to pumping, to gettering with filament

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material evaporation, and to plasma conditioning) that they do not hinder the observation of the ''pure'' sputtering effects at our ion flux densities of $\approx 10^{16}$ proton cm^{-2} s^{-1}.

4. Fine permeability variations at $E_k < 50$ eV

One can see that permeation, and particularly hydrogen compression by the membrane, still do not remain exactly constant when the ion energy varies over the range of $E_k < 50$ eV, and, correspondingly, the state of the surface (i.e., α_0) does not stay exactly the same as well. There are two opposite trends: the first one, exhibited at $E_k > 2$ eV, reveals an α_0 increase, while the other one that starts to prevail at $E_k > 7$ eV shows an α_0 drop. These peculiarities of behavior were quite reproducible, but their causes are not clear for the moment.

5. Membrane asymmetry

One can see on Fig. 7 that the asymmetry factor, α_L / α_0, at the plasma driven superpermeation reaches ≈ 10. The existence of this favorable asymmetry is revealed quite clearly by comparing the permeation and compression graphs (Fig. 6) and taking into account Eqs. (1) and (6); this was verified with a number of other independent methods. The asymmetry is only partly caused by the tantalum deposition onto the surfaces of the niobium (and vanadium) membranes. The favorable asymmetry (i.e., the decrease in α_0 at a constant α_L) mainly has its origin in the plasma operation, and particularly in the ion bombardment with an energy of 10–50 V. One can suppose that the bombardment with low energy hydrogen ions results in a modification of the monolayer impurity film, even though such ions are unable to sputter it. It may be noted that a similar effect was observed in ion beam experiments where chemically active gases (e.g., O_2 and H_2O) were continuously admitted at the upstream side in the course of ion bombardment: a favorable asymmetry was formed despite the ion sputtering.[2,7,17,21]

IV. CONCLUSION

Stable plasma driven superpermeation demanding no special efforts to maintain the required state of membrane surfaces (a spontaneous superpermeation) was observed on niobium and vanadium membranes at floating potential (i.e., at particle energies of up to a few eV) at all densities of permeation fluxes investigated up to 2×10^{17} atom/cm^2 s. The experiments with the biased niobium membrane showed that such a spontaneous superpermeation exists at ion energies of up to 60–70 eV. On the basis of the compression data, one can conclude that the state of the surface facing the plasma remains, in the first approximation, unchanged under bombardment with hydrogen ions with energies of up to ≈ 50 eV, and that this energy is, most probably, the threshold of physical sputtering of the nonmetallic (O, C) monolayer film responsible for the surface barrier. At the same time the chemical sputtering of this film does not seem to occur at all at energies below 50 eV. Thus superpermeability appears to be an intrinsic feature of the investigated system ''nonmetallic film–metal–hydrogen particles of an energy lower than 50 eV.''

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